Agentic Artificial Intelligence (Agentic AI)

Agentic Artificial Intelligence (Agentic AI) is an advanced AI approach that enables autonomous systems to perceive real-world environments, analyze information, make decisions, plan actions, and execute tasks to achieve specific objectives with minimal human intervention. Unlike conventional AI models that primarily focus on prediction and classification, Agentic AI integrates autonomous decision-making, reasoning, planning, memory, tool utilization, multi-agent collaboration, real-time data processing, and feedback-driven learning to solve complex real-world problems. Its potential applications include healthcare monitoring, intelligent transportation, industrial automation, predictive maintenance, disaster management, environmental monitoring, smart agriculture, and energy optimization. However, existing Agentic AI frameworks face several challenges, including unreliable decision-making, limited adaptability to dynamic environments, error propagation, inefficient coordination among agents, high computational overhead, inadequate explainability, and difficulties in ensuring safe and reliable autonomous operations. In this research, the working principles of conventional single-agent and multi-agent AI frameworks will first be investigated using publicly available real-world datasets and application-specific benchmark environments. Subsequently, potential improvements will be explored through adaptive decision-making, uncertainty-aware reasoning, dynamic task planning, explainable decision mechanisms, intelligent agent coordination, self-correction strategies, and computationally efficient agent architectures. The proposed modifications should be supported by mathematical formulations, theoretical proofs or derivations wherever applicable, and experimental validation. Their effectiveness will be compared with existing Agentic AI frameworks in terms of decision accuracy, task completion rate, response time, computational efficiency, robustness, adaptability, explainability, and reliability under realistic operating conditions.

Problem Statement 16:

To investigate the working principles, mathematical foundations, and practical limitations of conventional Agentic Artificial Intelligence (Agentic AI) frameworks for solving real-world problems and develop an enhanced, application-oriented Agentic AI system through novel mathematical or algorithmic modifications. The proposed framework should address challenges related to autonomous decision-making, real-time environmental adaptation, multi-agent coordination, uncertainty handling, inefficient task planning, error propagation, and computational complexity. The research should focus on practical applications such as healthcare monitoring, industrial predictive maintenance, intelligent transportation, disaster management, smart agriculture, or environmental monitoring using publicly available real-world datasets and realistic simulation environments. The proposed improvements should be supported by mathematical models, theoretical proofs, or experimental implementations and validated through comparative analysis with conventional AI, single-agent, and multi-agent frameworks. The effectiveness of the proposed system should be evaluated in terms of task completion rate, decision accuracy, response time, resource utilization, computational efficiency, robustness, explainability, safety, and generalization, with particular emphasis on developing reliable, adaptive, interpretable, and computationally efficient autonomous systems capable of addressing complex real-world challenges.

Machine Learning Project Evaluation Sheet:

S. No.	Evaluation Criteria	Evaluation Parameters	Max. Marks
1	Title Introduction and contribution	Problem clarity, motivation, contribution	5
2	Literature Review	Existing methods, limitations, research gaps	5
3	Dataset and Preprocessing	Dataset selection, EDA, preprocessing, feature engineering	5
4	Implementation of Proposed model	Algorithm correctness, implementation, hyperparameter tuning	15
5	Proposed Methodology , compare with baseline model, ablation study	Novelty, improvements, mathematical justification	10
6	Results , discussion and Comparative Analysis - table and graph	Performance metrics, baseline comparison, interpretation	10
7	Technical Viva + code demo	Conceptual understanding, mathematics, implementation	30
8	Presentation(10 slides) and Documentation(8 pages max)	Presentation quality, diagram demonstration, code and report	20
		TOTAL	**100**

Project status and progress discussion & presentation dates: 3rd, 10th and 17th Nov, 2026